Strain Energy Method

In a structure, material of which is linear elastic ($\sigma \propto \epsilon$, $P \propto \Delta$), temperature is constant, & supports are unyielding (supports should not settle), the deflection of any point in the direction of force (*) is equal to the 1st Partial derivative of total strain Energy with respect to the force.

This theorem is called Castigliano's 2nd Theorem.

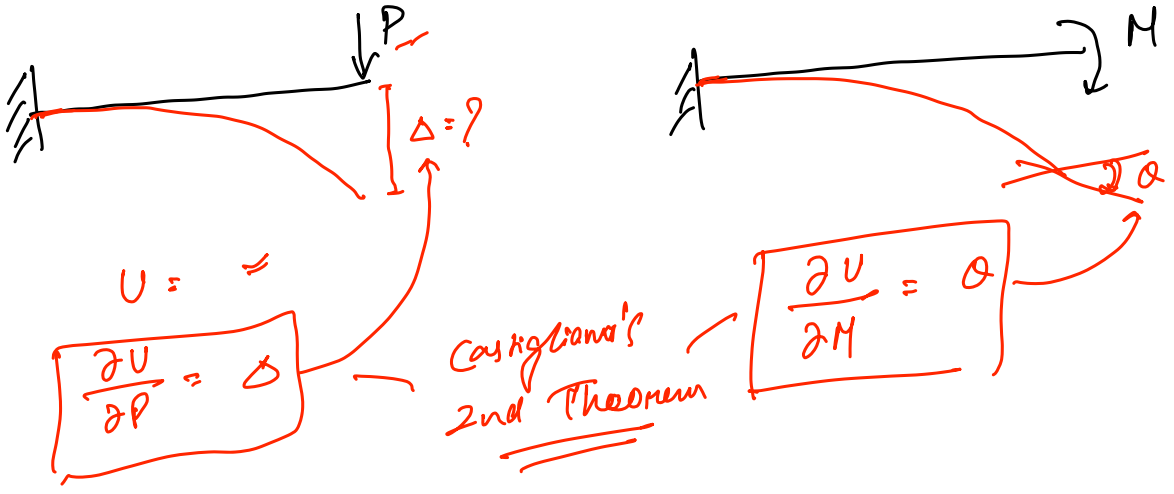
{ eg. $f(x,y) = 3x^2 + 4y + 6$

$\left(\frac{\partial f}{\partial x}\right) = 6x + 0 + 0$ | $\left(\frac{\partial f}{\partial y}\right) = 0 + 4 + 0$ }

If 'U' = Total S.E.

then, $\boxed{\frac{\partial U}{\partial P} = \Delta}$ ← Castigliano's 2nd Theorem.

{ $\boxed{\frac{\partial U}{\partial \Delta} = P}$ ← Castigliano's 1st Theorem }



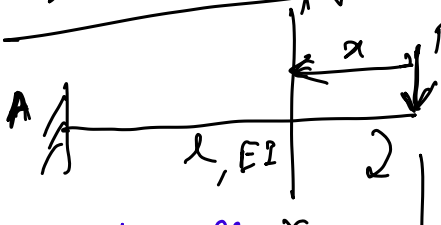
(*) This method is applicable for Both Determinate & Indeterminate structures.

If U = Total SE.

① Strain Energy due to Bending Moment

$$U = \int \frac{(M_x)^2 \, dx}{2EI}$$

$M_x \rightarrow$ BM at any section 'x'



$M_B = 0, \quad M_A = -Pl$

$M_x = (-Px)$

$$U = \int_{x=0}^{x=l} \frac{(-Px)^2 \, dx}{2EI}$$

$x=0 \rightarrow x=l$

$$\int x^n \, dx \quad \frac{x^{n+1}}{n+1}$$

$$\frac{d(x^n)}{dx} \quad nx^{n-1}$$

$$= \frac{P^2}{2EI} \int_0^l x^2 \, dx = \frac{P^2}{2EI} \frac{x^3}{3} \Big|_0^l$$

$$= \frac{P^2}{6EI} [l^3 - 0]$$

$$U = \frac{P^2 l^3}{6EI}$$

② Strain Energy due to Axial force (F)

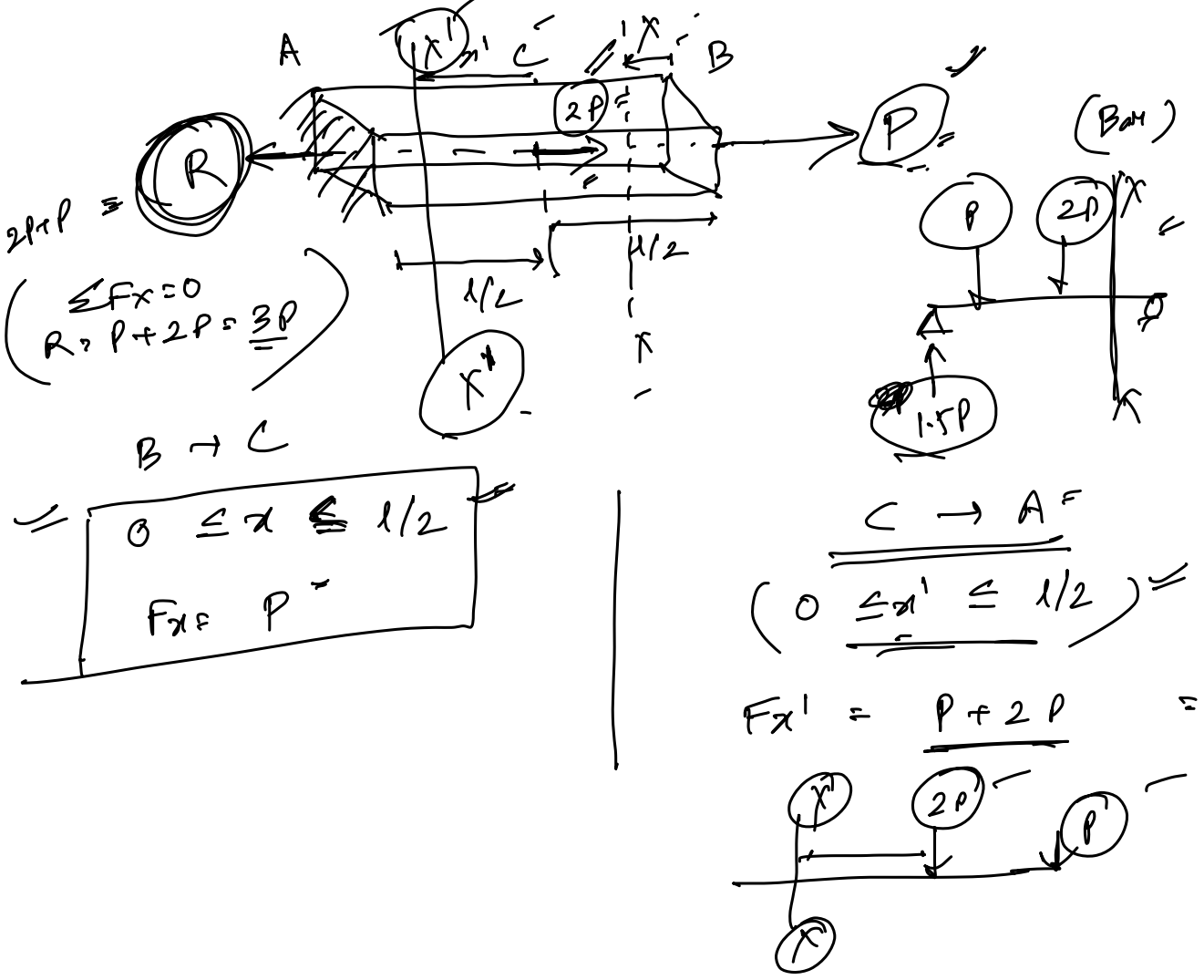
$$U = \int \frac{(F_x)^2 \, dx}{2AE}$$

$F_x \rightarrow$ Axial force at any section

$M_x = -Px$

P_x

$$\begin{Bmatrix} EI \\ AE \\ GI_p \end{Bmatrix} / L$$



$\Sigma F_x = 0$

$R = P + 2P = 3P$

$B \to C$

$0 \le x \le l/2$

$F_x = P$

$C \to A$

$0 \le x' \le l/2$

$F_{x'} = P + 2P = R$

$M_x =$

BC

$F_x = P$

CA

$F_x = 3P$

$$U = \int_0^{l} \frac{(F_x)^2 dx}{2AE}$$

U_{CB} + U_{CA}

$$= \int_0^{1/2} \frac{P^2 dx}{2AE} + \int_0^{1/2} \frac{(3P)^2 dx}{2AE} =$$

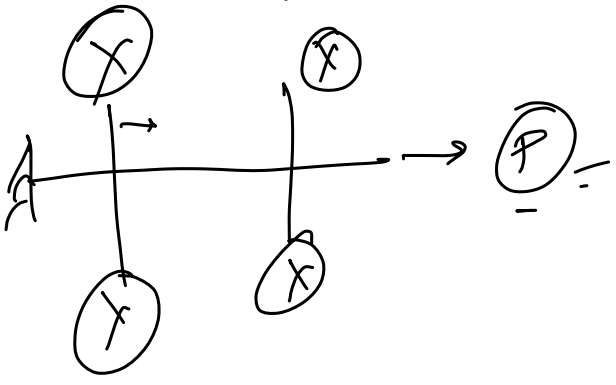
$$=$$

$$= \frac{P^2}{2AE}\int_0^{l/2} dx + \frac{9P^2}{2AE}\int_0^{l/2} dx$$

$$= \frac{P^2}{2AE}\frac{l}{2} + \frac{9P^2}{2AE}\frac{l}{2}$$

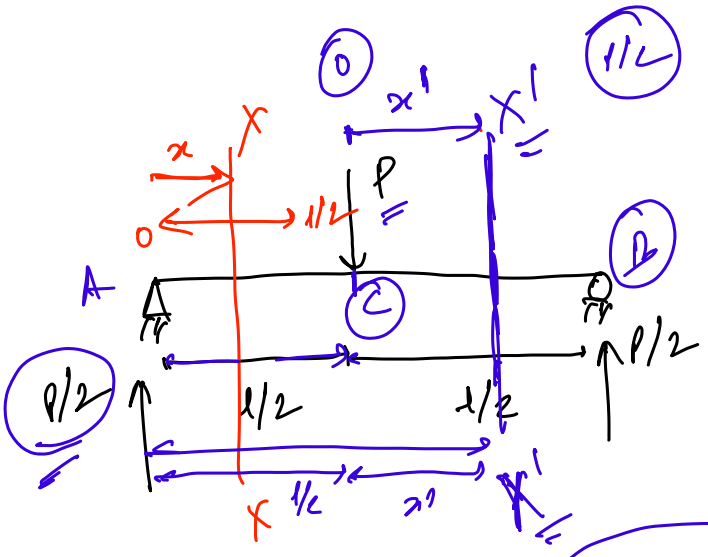
$$= \frac{P^2 l}{4AE} + \frac{9P^2 l}{4AE}$$

$$= \frac{10\,P^2 l}{4AE}$$



$$\Rightarrow U = \int_0^{l} \frac{P^2\,dx}{2AE}$$

$$= \frac{P^2 l}{2AE}$$



$U_{BM} = U_{AC} + U_{CB}$

$M_x = \frac{P}{2}x \quad (0 \le x \le l/2)$ [AC]

$M_x = \frac{P}{2}\left(\frac{l}{2} + x'\right) - Px'$ · [CB] $0 \le x' \le l/2$

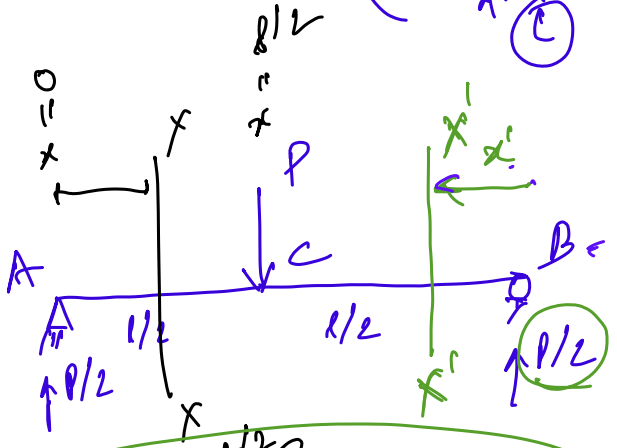
$$U_{AC} = \int_0^{l/2} \frac{\left(\frac{P}{2}x\right)^2 dx}{2EI}$$

(+)

$$U_{BC} = \int_{x'=0}^{x'=l/2} \frac{\left[\left(\frac{P}{2}\right)\left(\frac{l}{2}+x'\right) - Px'\right]^2 dx}{2EI}$$

C

$x'=0$ ← C



(+)

BC

$x'=0$ → B

$x'=l/2$ → C

$M_{x'} = \left(\frac{P}{2}x'\right)$

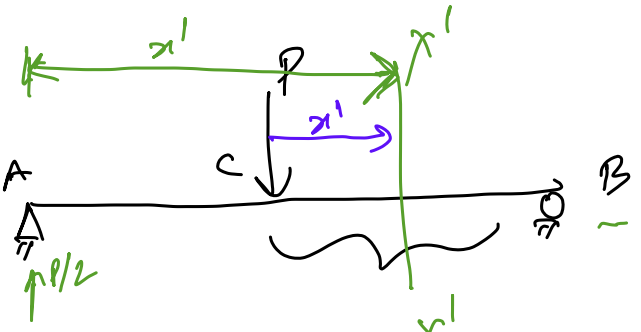
$$U_{AC} = \int_{x=0}^{x=l/2} \frac{\left(\frac{P}{2}x\right)^2 dx}{2EI}$$

$$U_{BC} = \int_{x'=0}^{x'=l/2} \frac{(M_{x'})^2 dx'}{2EI}$$

$x'=0$

$$U_{BC} = \int_{x'=0}^{x'=l/2=C} \frac{\left(\frac{P}{2}x'\right)^2 dx}{2EI}$$

$x'=0$ (B)



$M_{x'} = \left[\frac{P}{2}x' - P(x'-l/2)\right]$

U_{CB}:

$C \rightarrow x' = l/2$

$B \rightarrow x' = l$

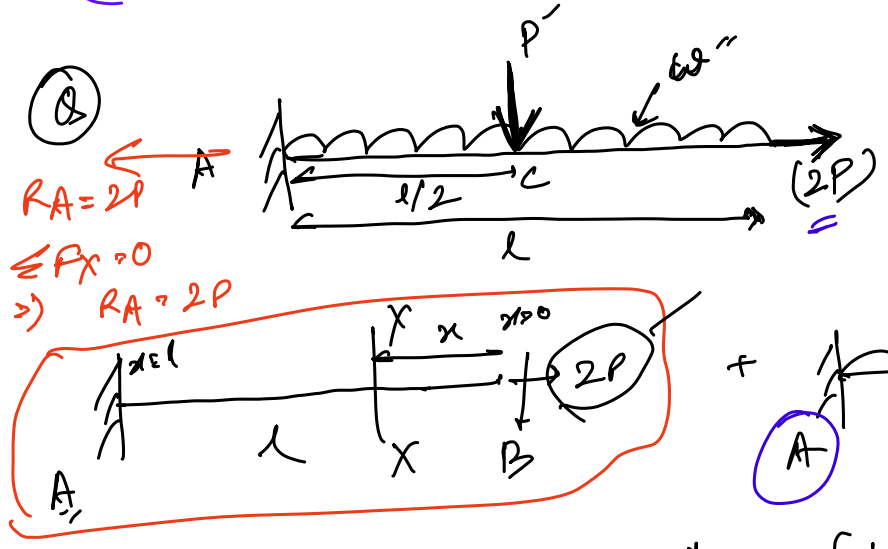
$= \int \frac{\left(\frac{P}{2}x' - P(x'-l/2)\right)^2 dx}{2EI}$ X

$x' = l/2$

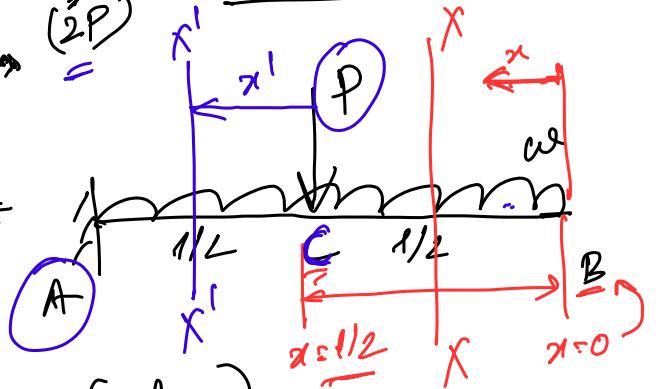
$C \rightarrow x' = 0$

$B \rightarrow x' = l/2$

$M_{x'} = \frac{P}{2}\left(\frac{l}{2} + x'\right) - Px'$



Find Total Strain Energy.



$+ \left(U_{AB}\right)$ Bending moment

$U = \left(U_{AB}\right)_{AF}$

$= \int_0^l \frac{(2P)^2 dx}{2AE}$

$= \frac{2P^2 l}{2AE}$

$= \left(\frac{2P^2 l}{AE}\right) =$

$\left(U_{AB}\right)_{BM}$

$= \left(U_{BC}\right) + U_{CA}$

$\left[M_{x=0} - \left(w x \frac{x}{2}\right) = \frac{-wx^2}{2}\right] \in$ FOR BC

$$U_{BC} = \int_{x=0}^{x=l/2} \frac{\left(\frac{-\omega x^2}{2}\right)^2 dx}{2EI}$$

$$U_{CA} =$$

$$M_{x'} = \left[\frac{-\omega\left(\frac{l}{2}+x'\right)^2}{2} - P x'\right]$$

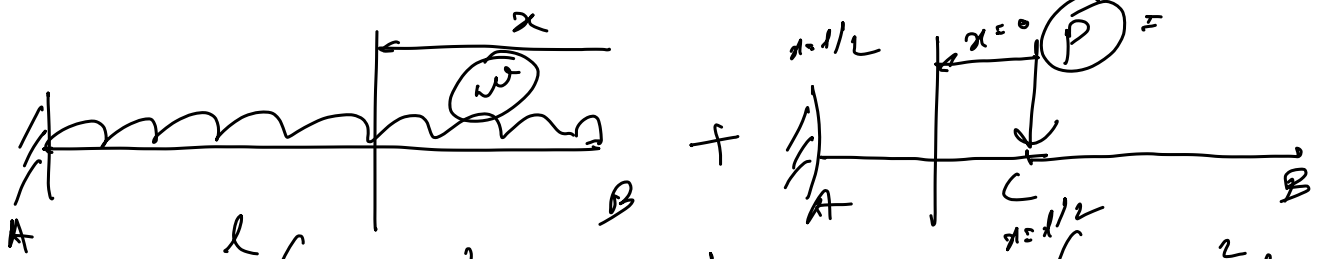
$x' = 0 \rightarrow C$

$x' = l/2 \rightarrow A$

$$U_{CA} = \int_{x'=0}^{x'=l/2} \frac{\left[\frac{-\omega\left(\frac{l}{2}+x'\right)^2}{2} - P x'\right]^2 dx}{2EI}$$

$$U = (U_{AB})_{AF} + (U_{AB})_{BM}$$

AM $\Rightarrow$ $$U = (U_{AB})_{AF} + (U_{BC})_{BM} + (U_{CA})_{BM}$$



$$U_{AB} = \int_0^l \frac{\left(\frac{-\omega x^2}{2}\right)^2 dx}{2EI}$$

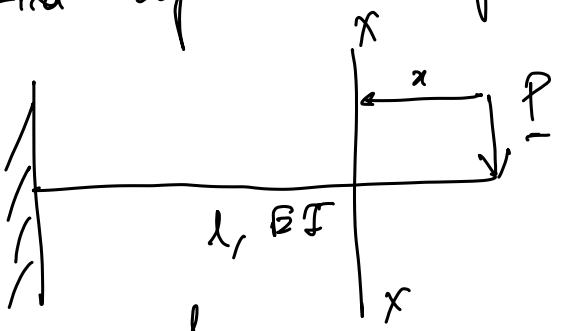
$$U_{AB} = \int_{x=0}^{x=l/2} \frac{(-Px)^2 dx}{2EI}$$

(3) Strain Energy due to Torsion

$$U = \int \frac{(T_x)^2 \, dx}{2 G I_p}$$

$T_x \rightarrow$ Torsion at any section $x-x$.

(Q) Find deflection at free end. (Using CE method)



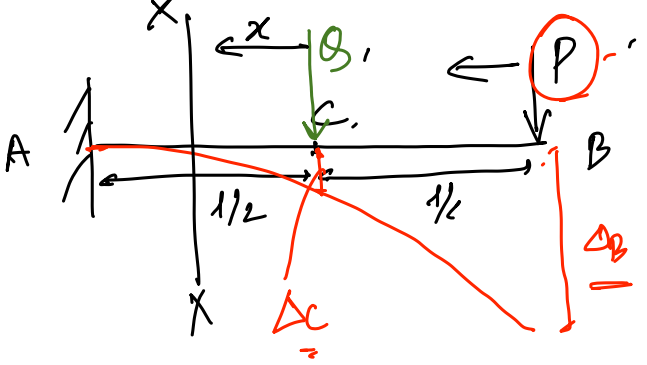
$M_x = (-Px)$

$$U = \int_0^l \frac{(-Px)^2 \, dx}{2EI} = \left(\frac{P^2 l^3}{6EI} \right)$$

$$\frac{\partial U}{\partial P} = \Delta = \frac{\partial}{\partial P} \left(\frac{P^2 l^3}{6EI} \right) = \frac{l^3}{6EI} \times 2P$$

$$= \frac{Pl^3}{3EI}$$

(8)



$\Delta_C = ?$ (Using SE Method)

Note: When deflection is required at any other point where force is not acting, then apply a Pseudo force in the direction of desired deflection & find the total SE due to given load & the applied Pseudo force & finally put the value of Pseudo force equal to zero.

U_{BC}

$M_x = (-Px)$

$$U_{BC} = \int_{x=0}^{x=l/2} \frac{(-Px)^2\,dx}{2EI}$$

$$U_{BC} = \frac{P^2}{2EI}\left(\frac{l}{2}\right)^3$$

$$U_{BC} = \frac{P^2 l^3}{16\,EI}$$

U_{CA}

$M_x = -Qx - P\left(\frac{l}{2}+x\right)$

$$U_{CA} = \int_{x=0}^{x=l/2} \frac{\left(-Qx - P\left(\frac{l}{2}+x\right)\right)^2 dx}{2EI}$$

$$U_{CA} = \frac{1}{2EI}\int \left(+Qx + \frac{Pl}{2} + Px\right)^2 dx$$

$$U_{CA} = \frac{1}{2EI}\int \left(Q^2x^2 + \frac{P^2l^2}{4} + P^2x^2 + 2Qx\,Px + Qx\frac{Pl}{2} + \frac{Q}{2}\frac{Pl}{2}\right) dx$$

$$\frac{\partial U_{BC}}{\partial Q} = 0$$

$$= \frac{1}{2EI}\int_0^{l/2}\left(Q^2x^2 + \frac{P^2l^2}{4} + P^2x^2 + 2QPx^2 + QPxl + P^2xl\right)$$

$$= \frac{1}{2EI}\left[\frac{Q^2x^3}{3} + \frac{P^2l^2}{4}x + \frac{P^2x^3}{3} + \frac{2QPx^3}{3} + \frac{QPlx^2}{2} + \frac{P^2lx^2}{2}\right]_0^{l/2}$$

$$U_{CA} = \frac{1}{2EI}\left[\frac{Q^2}{3}\frac{l^3}{8} + \frac{P^2l^2}{4}\frac{l}{2} + \frac{P^2}{3}\frac{l^3}{8} + \frac{2QP}{3}\frac{l^3}{8} + \frac{QPl}{8}l^2 + \frac{P^2l}{8}l^2\right]$$

$$U = U_{BC} + U_{CA} \qquad \Delta_C = ?$$

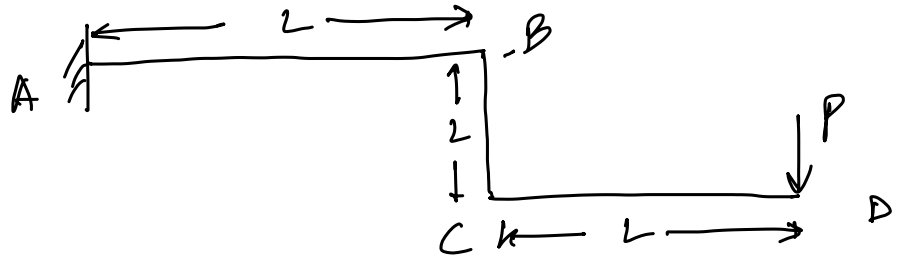
$$\left(\frac{\partial U}{\partial Q}\right)_{Q=0} = 0 + \left[\frac{2Ql^3}{24} + \frac{Pl^3}{12} + \frac{Pl^3}{8}\right]\frac{1}{2EI}$$

$$\left(\frac{\partial U}{\partial Q}\right) = \frac{1}{2EI}\left(\frac{Pl^3}{12} + \frac{Pl^3}{8} + \frac{Ql^3}{12}\right)$$

$$\Delta_C = \left(\frac{\partial U}{\partial Q}\right) = \frac{1}{2EI}\left[\frac{5Pl^3}{24} + \frac{Ql^3}{12}\right] \qquad (Q=0)$$

$$\boxed{\Delta_C = \frac{5Pl^3}{48EI}}$$

(Q)

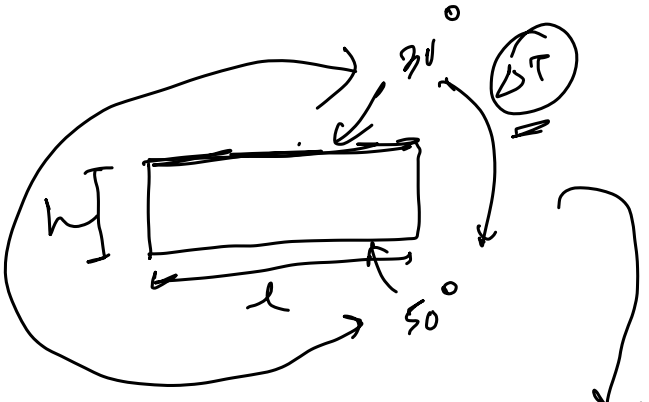


Calculate deflection at 'D' & 'B'.

Sunday, Monday Tuesday | Wednesday → Simple

= Properties of Material

{ . . . . . } { . . . . . }



$T_0 = 25°$

$\frac{M}{EI} = \frac{1}{R} = \frac{d^2y}{dx^2}$



$\Delta L \propto L \Delta T$

$$\Delta_{midspan} = \frac{\alpha (\Delta T) l^2}{8h}$$

$\alpha \rightarrow$ Coeff of Thermal Expansion
$l \rightarrow$ Span
$h \rightarrow$ Depth of the Beam

$\Delta T \rightarrow$ Diff in Temperature b/w Top & Bottom face of the Beam.